

- 9 An aqueous mixture of ethanoic acid, $\text{CH}_3\text{CO}_2\text{H}$, and sodium ethanoate, $\text{CH}_3\text{CO}_2^-\text{Na}^+$, behaves as a buffer. This solution maintains an almost constant pH when small amounts of either H^+ ions or OH^- ions are added.

Which equation represents one of the reactions that occurs?

- A $\text{CH}_3\text{CO}_2\text{H} + \text{H}^+ \rightarrow \text{H}_2\text{O} + \text{CH}_3\text{CO}_2^-$
- B $\text{CH}_3\text{CO}_2\text{H} + \text{OH}^- \rightarrow \text{H}_2\text{O} + \text{CH}_3\text{CO}_2^-$
- C $\text{CH}_3\text{CO}_2^-\text{Na}^+ + \text{H}^+ \rightarrow \text{NaH} + \text{CH}_3\text{CO}_2^-$
- D $\text{CH}_3\text{CO}_2^-\text{Na}^+ + \text{OH}^- \rightarrow \text{NaOH} + \text{CH}_3\text{CO}_2^-$